

# UNIVERSITY OF ENGINEERING AND TECHNOLOGY PESHAWAR, PAKISTAN

Main Campus

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## DATA STRUCTURES

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### Lab 01 Assignment

**Submitted By**

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Semester: 2nd Semester, BS Computer Science

Section: B

Submission Date: 14-09-2026

**Submitted To: Sir Mudasir**

**DEPARTMENT OF COMPUTER SCIENCE**

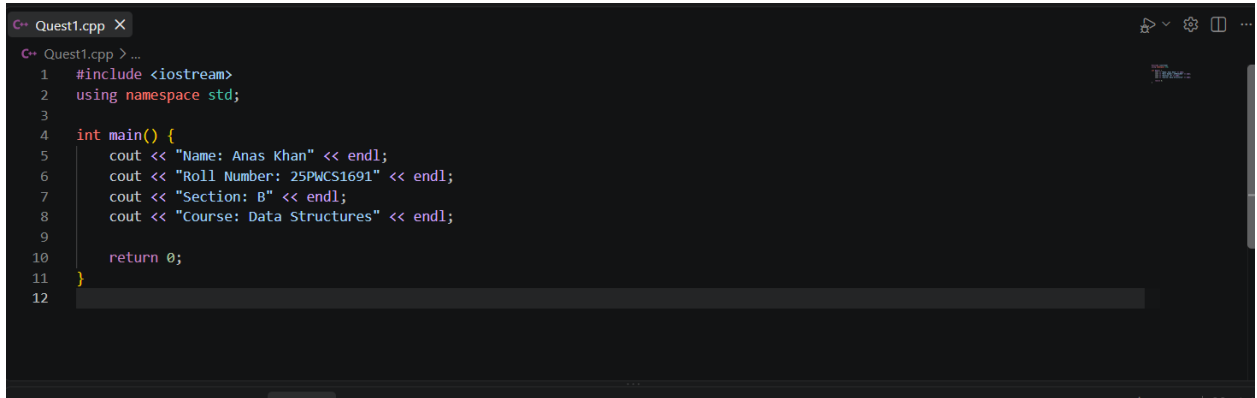
UNIVERSITY OF ENGINEERING AND TECHNOLOGY, PESHAWAR, PAKISTAN

## Lab Exercises 1.1

### Task 1:

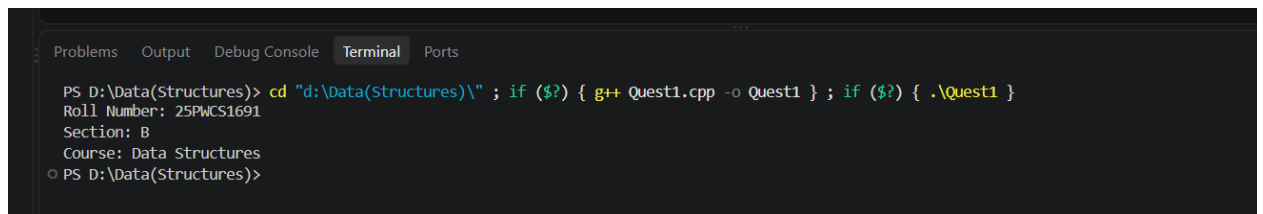
Modify the program to print your name, roll number, and course.

### Program:



```
Quest1.cpp X
C++ Quest1.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      cout << "Name: Anas Khan" << endl;
6      cout << "Roll Number: 25PWCS1691" << endl;
7      cout << "Section: B" << endl;
8      cout << "Course: Data Structures" << endl;
9
10     return 0;
11 }
12
```

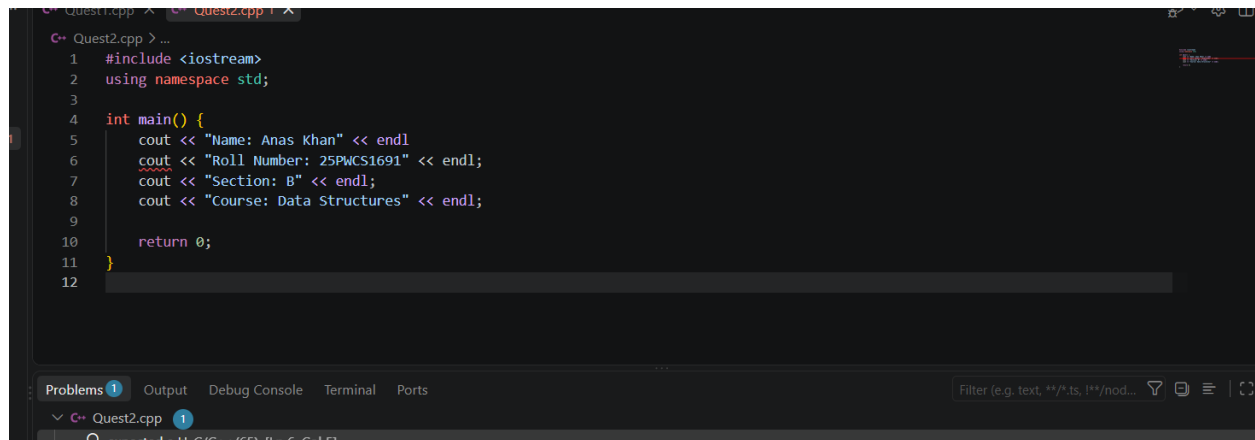
### Output:



```
Problems Output Debug Console Terminal Ports
PS D:\Data(Structures)> cd "d:\Data(Structures)\\" ; if ($?) { g++ Quest1.cpp -o Quest1 } ; if ($?) { .\Quest1 }
Roll Number: 25PWCS1691
Section: B
Course: Data Structures
PS D:\Data(Structures)>
```

2- Introduce a deliberate syntax error (remove a semicolon) and observe the compiler error message.

Program:

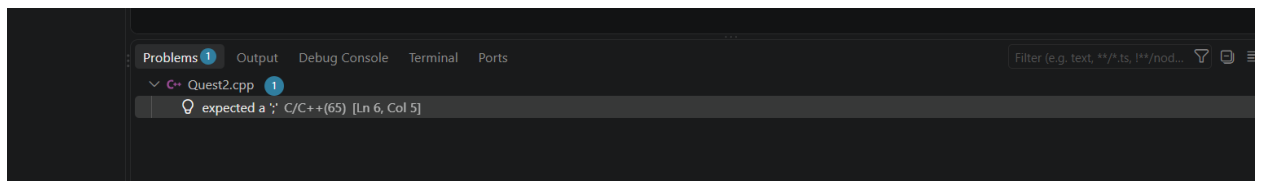


The screenshot shows a C++ code editor with a file named Quest2.cpp. The code is as follows:

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     cout << "Name: Anas Khan" << endl;
6     cout << "Roll Number: 25PWCS1691" << endl;
7     cout << "Section: B" << endl;
8     cout << "Course: Data Structures" << endl;
9
10    return 0;
11 }
12
```

The bottom panel of the IDE shows the 'Problems' tab with one error listed: 'expected a ';' C/C++(65) [Ln 6, Col 5]'. This error points to the line where the roll number is printed, indicating a missing semicolon.

Output:



The screenshot shows the 'Problems' tab in the IDE, displaying the same error message as before: 'expected a ';' C/C++(65) [Ln 6, Col 5]'. The error is highlighted with a light blue background.

## Lab Exercises 1.2

### Task 1:

Write a program to take a student's marks in 3 subjects and print total and average.

### Program:

```
Quest3.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      float subject1, subject2, subject3;
6      float total, average;
7
8      cout << "Enter marks for Subject 1: ";
9      cin >> subject1;
10
11     cout << "Enter marks for Subject 2: ";
12     cin >> subject2;
13
14     cout << "Enter marks for Subject 3: ";
15     cin >> subject3;
16
17     total = subject1 + subject2 + subject3;
18     average = total / 3;
19
20     cout << "Total Marks: " << total << endl;
21     cout << "Average Marks: " << average << endl;
22
23     return 0;
24 }
```

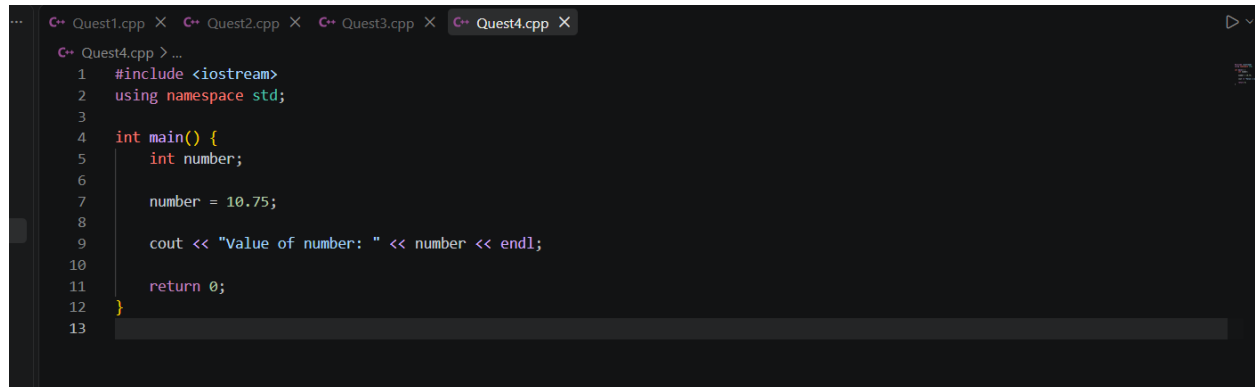
### Output:

```
11     cout << "Enter marks for Subject 2: ";
12     cin >> subject2;
...
Problems 1 Output Debug Console Terminal Ports
• PS D:\Data(Structures)> cd "d:\Data(Structures)" ; if ($?) { g++ Quest3.cpp -o Quest3 } ; if ($?) { .\Quest3 }
Enter marks for Subject 1: 10
Enter marks for Subject 2: 20
Enter marks for Subject 3: 30
Total Marks: 60
Average Marks: 20
○ PS D:\Data(Structures)> |
```

## Task 2 :

Explore what happens when you assign a float value to an int variable (type narrowing).

## Program:



```
C++ Quest1.cpp X C++ Quest2.cpp X C++ Quest3.cpp X C++ Quest4.cpp X
C++ Quest4.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int number;
6
7      number = 10.75;
8
9      cout << "Value of number: " << number << endl;
10
11     return 0;
12 }
13
```

## Output:



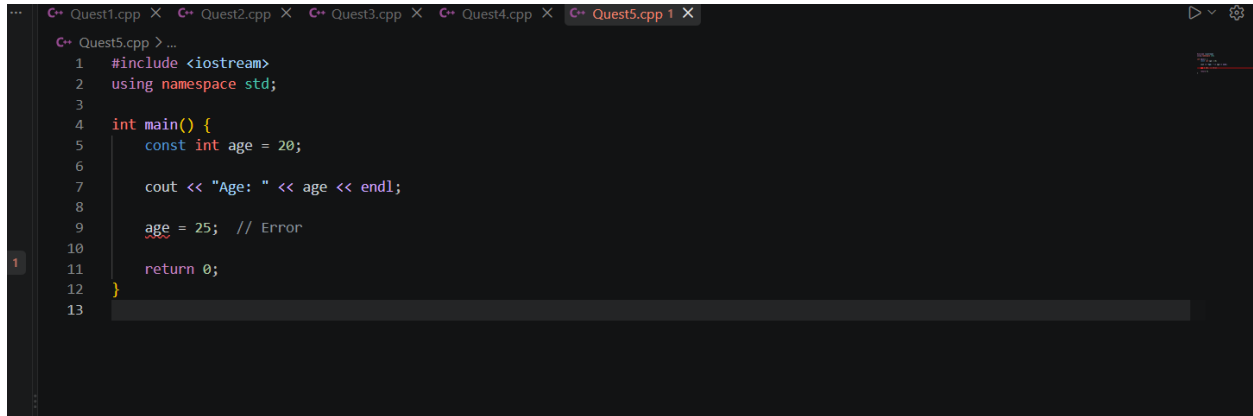
```
Problems Output Debug Console Terminal Ports
PS D:\Data(Structures)> cd "d:\Data(Structures)\\" ; if ($?) { g++ Quest4.cpp -o Quest4 } ; if ($?) { .\Quest4 }
Value of number: 10
PS D:\Data(Structures)>
```

10.75 is converted to 10. The .75 is lost because an int can store only whole numbers.

### Task 3 :

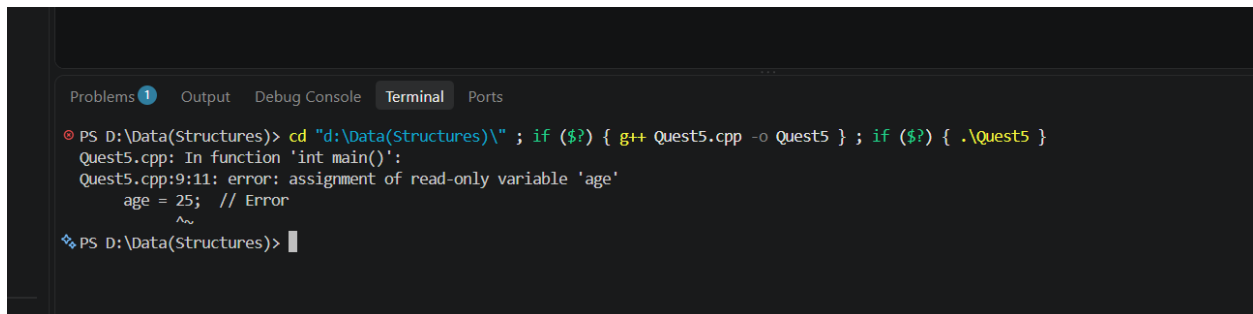
Use const to declare a constant and try modifying it — observe the compiler error.

### Program:



```
C++ Quest1.cpp x C++ Quest2.cpp x C++ Quest3.cpp x C++ Quest4.cpp x C++ Quest5.cpp 1 x
C++ Quest5.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      const int age = 20;
6
7      cout << "Age: " << age << endl;
8
9      age = 25; // Error
10
11     return 0;
12 }
13
```

### Output:



```
Problems 1 Output Debug Console Terminal Ports
PS D:\Data(Structures)> cd "d:\Data(Structures)\\" ; if ($?) { g++ Quest5.cpp -o Quest5 } ; if ($?) { .\Quest5 }
Quest5.cpp: In function 'int main()':
Quest5.cpp:9:11: error: assignment of read-only variable 'age'
    age = 25; // Error
    ^~
PS D:\Data(Structures)>
```

Because age was declared using const, its value cannot be modified.

## Lab Exercises 1.3

### Task 1:

Write a program to swap two numbers without using a third variable.

### Program:

```
C++ Quest6.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int a, b;
6
7      cout << "Enter first number: ";
8      cin >> a;
9
10     cout << "Enter second number: ";
11     cin >> b;
12
13     cout << "Before swapping:" << endl;
14     cout << "a = " << a << ", b = " << b << endl;
15
16     a = a + b;
17     b = a - b;
18     a = a - b;
19
20     cout << "After swapping:" << endl;
21     cout << "a = " << a << ", b = " << b << endl;
22
23     return 0;
24 }
25
```

### Output:

```
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16     a = a + b;
17     b = a - b;
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```

## Program:

```
Quest7.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int number;
6
7      cout << "Enter a number: ";
8      cin >> number;
9
10     if (number % 2 == 0) {
11         cout << "The number is even." << endl;
12     } else {
13         cout << "The number is odd." << endl;
14     }
15
16     return 0;
17 }
18
```

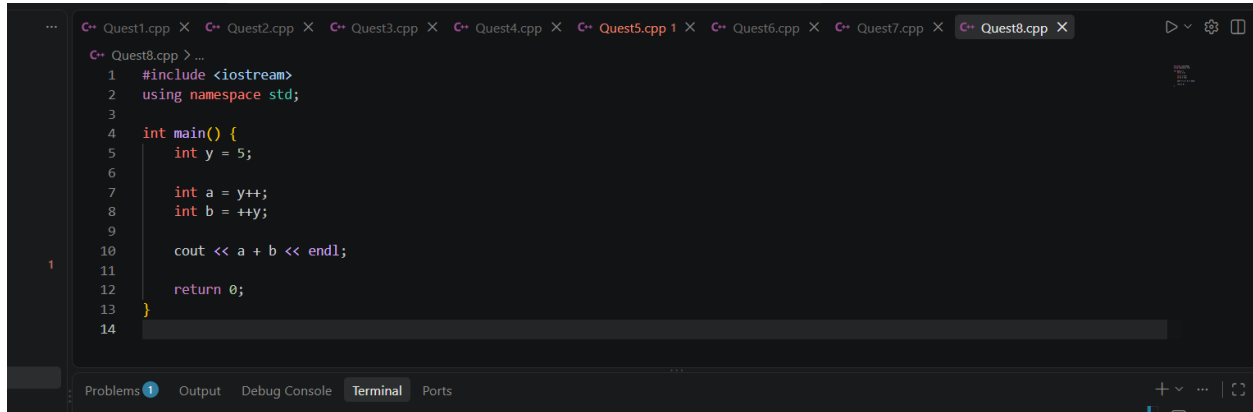
## Output:

```
15
16     return 0;
Problems 1 Output Debug Console Terminal Ports
• PS D:\Data(Structures)> cd "d:\Data(Structures)\\" ; if ($?) { g++ Quest7.cpp -o Quest7 } ; if ($?) { .\Quest7 }
Enter a number: 10
The number is even.
• PS D:\Data(Structures)> cd "d:\Data(Structures)\\" ; if ($?) { g++ Quest7.cpp -o Quest7 } ; if ($?) { .\Quest7 }
Enter a number: 5
The number is odd.
○ PS D:\Data(Structures)> |
```

### Task 3:

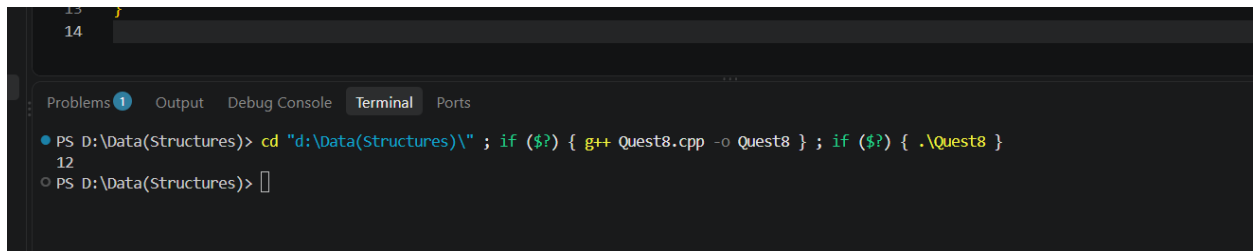
Predict and verify the output of `int y = 5; cout << y++ + ++y;`

### Program:



```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int y = 5;
6
7     int a = y++;
8     int b = ++y;
9
10    cout << a + b << endl;
11
12    return 0;
13 }
14
```

### Output:



```
PS D:\Data(Structures)> cd "d:\Data(Structures)" ; if ($?) { g++ Quest8.cpp -o Quest8 } ; if ($?) { .\Quest8 }
12
PS D:\Data(Structures)>
```

## Lab Exercises 1.4

### Task 1:

Write a program to check if a year is a leap year.

### Program:

```
C++ Quest9.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int year;
6
7      cout << "Enter a year: ";
8      cin >> year;
9
10     if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0)) {
11         cout << year << " is a leap year." << endl;
12     } else {
13         cout << year << " is not a leap year." << endl;
14     }
15
16     return 0;
17 }
18
```

## Output:

```
16     return 0;
17 }
18
Problems 1 Output Debug Console Terminal Ports
• PS D:\Data(Structures)> cd "d:\Data(Structures)\\" ; if ($?) { g++ Quest9.cpp -o Quest9 } ; if ($?) { .\Quest9 }
Enter a year: 2005
2005 is not a leap year.
○ PS D:\Data(Structures)>
```

## Task 2:

Build a simple traffic light simulator using switch (Red/Yellow/Green → message).

## Program:

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int choice;
6
7     cout << "Traffic Light Simulator" << endl;
8     cout << "1. Red" << endl;
9     cout << "2. Yellow" << endl;
10    cout << "3. Green" << endl;
11
12    cout << "Enter your choice: ";
13    cin >> choice;
14
15    switch (choice) {
16        case 1:
17            cout << "Red: Stop!" << endl;
18            break;
19
20        case 2:
21            cout << "Yellow: Get Ready!" << endl;
22            break;
23
24        case 3:
25            cout << "Green: Go!" << endl;
26            break;
27
28        default:
29            cout << "Invalid choice!" << endl;
30    }
31
32    return 0;
33 }
```

## Output:

```
PS D:\Data(Structures)> cd "d:\Data(Structures)\\" ; if ($?) { g++ Quest10.cpp -o Quest10 } ; if ($?) { .\Quest10 }
Traffic Light Simulator
1. Red
2. Yellow
3. Green
Enter your choice: 3
Green: Go!
PS D:\Data(Structures)> 2
2
PS D:\Data(Structures)> cd "d:\Data(Structures)\\" ; if ($?) { g++ Quest10.cpp -o Quest10 } ; if ($?) { .\Quest10 }
Traffic Light Simulator
1. Red
2. Yellow
3. Green
Enter your choice: 1
Red: Stop!
PS D:\Data(Structures)> 
```

## Task 3:

Rewrite Step 1's grade calculator using nested ternary operators.

## Program:

```
C++ Quest11.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int marks;
6      char grade;
7
8      cout << "Enter your marks: ";
9      cin >> marks;
10
11     grade = (marks >= 90) ? 'A' :
12             (marks >= 80) ? 'B' :
13             (marks >= 70) ? 'C' :
14             (marks >= 60) ? 'D' :
15             'F';
16
17     cout << "Grade: " << grade << endl;
18
19     return 0;
20 }
21
```

## Output:

```
Problems 1 Output Debug Console Terminal Ports
• PS D:\Data(Structures)> cd "d:\Data(Structures)" ; if ($?) { g++ Quest11.cpp -o Quest11 } ; if ($?) { .\Quest11 }
Enter your marks: 24
Grade: F
• PS D:\Data(Structures)> cd "d:\Data(Structures)" ; if ($?) { g++ Quest11.cpp -o Quest11 } ; if ($?) { .\Quest11 }
Enter your marks: 90
Grade: A
○ PS D:\Data(Structures)> |
```

## Lab Exercises 1.5

### Task 1:

Print a right-angled triangle pattern of stars using nested loops.

Program:

```
1.cpp × C++ Quest5.cpp 1 × C++ Quest6.cpp × C++ Quest7.cpp × C++ Quest8.cpp × C++ Quest9.cpp × C++ Quest10.cpp × C++ Quest11.cpp × C++ Quest12.cpp ×
C++ Quest12.cpp > ...
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int rows;
6
7     cout << "Enter number of rows: ";
8     cin >> rows;
9
10    for (int i = 1; i <= rows; i++) {
11        for (int j = 1; j <= i; j++) {
12            cout << "* ";
13        }
14        cout << endl;
15    }
16
17    return 0;
18 }
19
```

Output:



```
Problems Output Debug Console Terminal Ports
PS D:\Data(Structures)> cd "d:\Data(Structures)\\" ; if ($?) { g++ Quest12.cpp -o Quest12 } ; if ($?) { .\Quest12 }
Enter number of rows: 5
*
* *
* * *
* * * *
* * * * *
```

### Task 2:

Check whether a number is prime using a for loop and break.

## Program:

```
cpp 1 X C++ Quest6.cpp X C++ Quest7.cpp X C++ Quest8.cpp X C++ Quest9.cpp X C++ Quest10.cpp X C++ Quest11.cpp X C++ Quest12.cpp X C++ Quest13.cpp X
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int number;
6     bool isPrime = true;
7
8     cout << "Enter a number: ";
9     cin >> number;
10
11     if (number <= 1) {
12         isPrime = false;
13     } else {
14         for (int i = 2; i < number; i++) {
15             if (number % i == 0) {
16                 isPrime = false;
17                 break;
18             }
19         }
20     }
21
22     if (isPrime) {
23         cout << number << " is a prime number." << endl;
24     } else {
25         cout << number << " is not a prime number." << endl;
26     }
27
28     return 0;
29 }
```

## Output:

```
13     } else {
14         for (int i = 2; i < number; i++) {
Problems Output Debug Console Terminal Ports
• PS D:\Data(Structures)> cd "d:\Data(Structures)\\" ; if ($?) { g++ Quest13.cpp -o Quest13 } ; if ($?) { .\Quest13 }
Enter a number: 201
201 is not a prime number.
• PS D:\Data(Structures)> cd "d:\Data(Structures)\\" ; if ($?) { g++ Quest13.cpp -o Quest13 } ; if ($?) { .\Quest13 }
Enter a number: 200
200 is not a prime number.
• PS D:\Data(Structures)> |
```

## Task 3:

Print only even numbers from 1–50, skipping odd numbers using continue.

## Program:

```
C++ Quest14.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      for (int i = 1; i <= 50; i++) {
7          if (i % 2 != 0) {
8              continue;
9          }
10
11          cout << i << " ";
12      }
13
14      return 0;
15 }
16
```

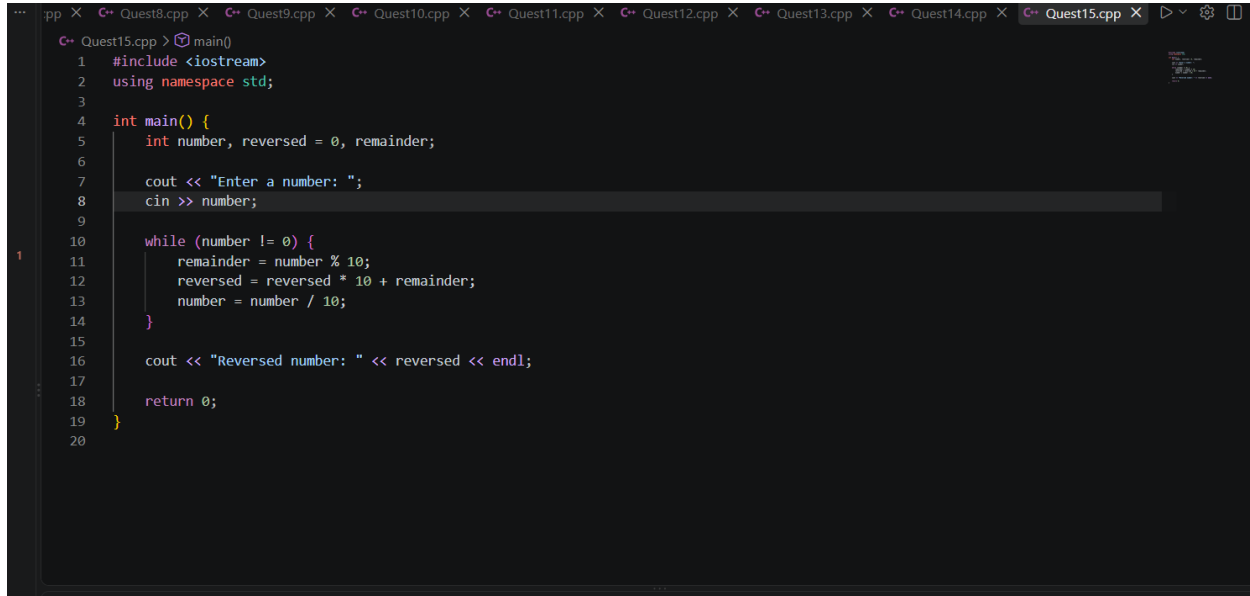
## Output:

```
Problems 1 Output Debug Console Terminal Ports
• PS D:\Data\Structures> cd "d:\Data\Structures\" ; if ($?) { g++ Quest14.cpp -o Quest14 } ; if ($?) { .\Quest14 }
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50
○ PS D:\Data\Structures>
```

#### Task 4:

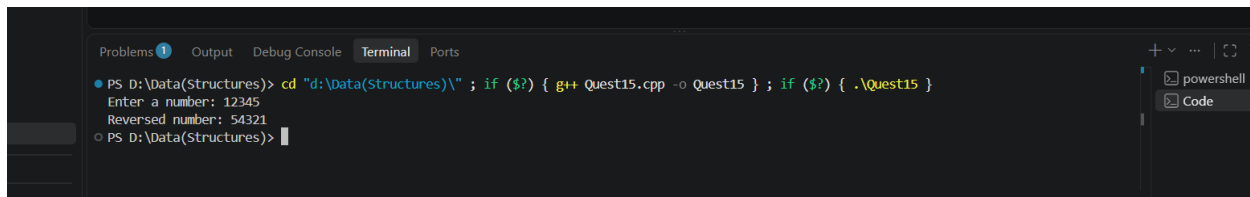
Reverse a number using a loop (e.g., 1234 → 4321).

#### Program:



```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int number, reversed = 0, remainder;
6
7      cout << "Enter a number: ";
8      cin >> number;
9
10     while (number != 0) {
11         remainder = number % 10;
12         reversed = reversed * 10 + remainder;
13         number = number / 10;
14     }
15
16     cout << "Reversed number: " << reversed << endl;
17
18     return 0;
19 }
20
```

#### Output:



```
PS D:\Data\Structures> cd "d:\Data\Structures\" ; if ($?) { g++ Quest15.cpp -o Quest15 } ; if ($?) { .\Quest15 }
Enter a number: 12345
Reversed number: 54321
PS D:\Data\Structures>
```

## Lab Exercises 1.6

### Task 1:

Write a recursive function to compute factorial of a number.

### Program:

```
C++ Quest16.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int factorial(int n) {
5      if (n == 0 || n == 1) {
6          return 1;
7      }
8
9      return n * factorial(n - 1);
10 }
11
12 int main() {
13     int number;
14
15     cout << "Enter a number: ";
16     cin >> number;
17
18     cout << "Factorial = " << factorial(number) << endl;
19
20     return 0;
21 }
22
```

### Output:

```
Problems 1 Output Debug Console Terminal Ports
• PS D:\Data\Structures> cd "d:\Data\Structures\" ; if ($?) { g++ Quest16.cpp -o Quest16 } ; if ($?) { .\Quest16 }
Enter a number: 3
Factorial = 6
○ PS D:\Data\Structures> |
```

### Task 2:

Write a function isPrime(int n) returning bool, and use it to print all primes between 1–100.

## Program:

```
C++ Quest17.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  bool isPrime(int n) {
5      if (n < 2) {
6          return false;
7      }
8
9      for (int i = 2; i * i <= n; i++) {
10         if (n % i == 0) {
11             return false;
12         }
13     }
14
15     return true;
16 }
17
18 int main() {
19     cout << "Prime numbers between 1 and 100:" << endl;
20
21     for (int i = 1; i <= 100; i++) {
22         if (isPrime(i)) {
23             cout << i << " ";
24         }
25     }
26
27     return 0;
28 }
29
```

## Output:

```
14
15     return true;
16 }

Problems 1 Output Debug Console Terminal Ports

PS D:\Data\Structures> cd "d:\Data\Structures\" ; if ($?) { g++ Quest17.cpp -o Quest17 } ; if ($?) { .\Quest17 }
Prime numbers between 1 and 100:
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97
PS D:\Data\Structures> cd "d:\Data\Structures\" ; if ($?) { g++ Quest17.cpp -o Quest17 } ; if ($?) { .\Quest17 }
Prime numbers between 1 and 100:
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97
PS D:\Data\Structures>
```

## Task 3:

Write an overloaded area() function for a circle (1 param), rectangle (2 params), and triangle (2 params, different formula) — think about how you'd disambiguate.

## Program:

```
C++ Quest18.cpp > ...
10  int area(int length, int width) {
13
14  // Area of triangle
15  double area(double base, double height) {
16      return 0.5 * base * height;
17  }
18
19  int main() {
20      double radius = 5;
21      int length = 10, width = 6;
22      double base = 8, height = 4;
23
24      cout << "Area of Circle = " << area(radius) << endl;
25      cout << "Area of Rectangle = " << area(length, width) << endl;
26      cout << "Area of Triangle = " << area(base, height) << endl;
27
28      return 0;
29  }
30
```

## Output:

```
Problems 1 Output Debug Console Terminal Ports
PS D:\Data(Structures)> cd "d:\Data(Structures)\\" ; if ($?) { g++ Quest18.cpp -o Quest18 } ; if ($?) { .\Quest18 }
Area of Circle = 78.5397
Area of Rectangle = 60
Area of Triangle = 16
PS D:\Data(Structures)>
```

## Task 4:

Write a function with a default argument, e.g., void greet(string name = "Guest").

## Program:

```
C++ Quest19.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  void greet(string name = "Guest") {
5      cout << "Hello, " << name << "!" << endl;
6  }
7
8  int main() {
9      greet();
10     greet("Anas");
11
12     return 0;
13 }
14
```

## Output:

```
Problems 1 Output Debug Console Terminal Ports
• PS D:\Data(Structures)> cd "d:\Data(Structures)" ; if ($?) { g++ Quest19.cpp -o Quest19 } ; if ($?) { .\Quest19 }
Hello, Guest!
Hello, Anas!
○ PS D:\Data(Structures)> |
```

## Lab Exercises 1.7

### Task 1:

Implement linear search on an array.

## Program:

```
C++ Quest20.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int arr[5] = {10, 20, 30, 40, 50};
6      int search, position = -1;
7
8      cout << "Enter number to search: ";
9      cin >> search;
10
11     for (int i = 0; i < 5; i++) {
12         if (arr[i] == search) {
13             position = i;
14             break;
15         }
16     }
17
18     if (position != -1) {
19         cout << "Number found at index: " << position << endl;
20     } else {
21         cout << "Number not found." << endl;
22     }
23
24     return 0;
25 }
26
```

## Output:

```
Problems  Output  Debug Console  Terminal  Ports
• PS D:\Data(Structures)> cd "d:\Data(Structures)" ; if ($?) { g++ Quest20.cpp -o Quest20 } ; if ($?) { .\Quest20.exe }
Enter number to search: 5
Number not found.
• PS D:\Data(Structures)> cd "d:\Data(Structures)" ; if ($?) { g++ Quest20.cpp -o Quest20 } ; if ($?) { .\Quest20.exe }
Enter number to search: 10
Number found at index: 0
○ PS D:\Data(Structures)>
```

## Task 2:

Implement bubble sort on an array of integers.

## Program:

```
C++ Quest20.cpp X C++ Quest21.cpp X
C++ Quest21.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int arr[5] = {50, 20, 40, 10, 30};
6
7      for (int i = 0; i < 5 - 1; i++) {
8          for (int j = 0; j < 5 - i - 1; j++) {
9              if (arr[j] > arr[j + 1]) {
10                 int temp = arr[j];
11                 arr[j] = arr[j + 1];
12                 arr[j + 1] = temp;
13             }
14         }
15     }
16
17     cout << "Sorted array: ";
18
19     for (int i = 0; i < 5; i++) {
20         cout << arr[i] << " ";
21     }
22
23     return 0;
24 }
25
```

## Output:

```
12  arr[j + 1] = temp;
Problems Output Debug Console Terminal Ports
● PS D:\Data(Structures)> cd "d:\Data(Structures)" ; if ($?) { g++ Quest21.cpp -o Quest21 } ; if ($?) { .\Quest21 }
Sorted array: 10 20 30 40 50
○ PS D:\Data(Structures)>
```

## Task 3:

Write a program to find the transpose of a matrix.

## Program:

```
C++ Quest20.cpp X C++ Quest21.cpp X C++ Quest22.cpp X
C++ Quest22.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int matrix[2][3] = {
6          {1, 2, 3},
7          {4, 5, 6}
8      };
9
10     cout << "Original Matrix:" << endl;
11
12     for (int i = 0; i < 2; i++) {
13         for (int j = 0; j < 3; j++) {
14             cout << matrix[i][j] << " ";
15         }
16         cout << endl;
17     }
18
19     cout << "Transpose of Matrix:" << endl;
20
21     for (int j = 0; j < 3; j++) {
22         for (int i = 0; i < 2; i++) {
23             cout << matrix[i][j] << " ";
24         }
25         cout << endl;
26     }
27
28     return 0;
29 }
30
```

## Output:

```
14     cout << matrix[i][j] << " ";
Problems Output Debug Console Terminal Ports
PS D:\Data(Structures)> cd "d:\Data(Structures)" ; if ($?) { g++ Quest22.cpp -o Quest22 } ; if ($?) { .\Quest22 }
3 6
PS D:\Data(Structures)> 
```

## Task 4:

Reverse an array in place (without using a second array).

## Program:

```
C++ Quest20.cpp X C++ Quest21.cpp X C++ Quest22.cpp X C++ Quest23.cpp X
C++ Quest23.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int arr[5] = {10, 20, 30, 40, 50};
6
7      int start = 0;
8      int end = 4;
9
10     while (start < end) {
11         int temp = arr[start];
12         arr[start] = arr[end];
13         arr[end] = temp;
14
15         start++;
16         end--;
17     }
18
19     cout << "Reversed array: ";
20
21     for (int i = 0; i < 5; i++) {
22         cout << arr[i] << " ";
23     }
24
25     return 0;
26 }
27
```

## Output:

```
26 }
27
Problems Output Debug Console Terminal Ports
● PS D:\Data(Structures)> cd "d:\Data(Structures)\\" ; if ($?) { g++ Quest23.cpp -o Quest23 } ; if ($?) { .\Quest23 }
Reversed array: 50 40 30 20 10
○ PS D:\Data(Structures)> █
```

## Lab Exercises 1.8

### Task 1,2,3, 4:

Count vowels and consonants in a given string.

Write a program to count the number of words in a sentence.

Implement your own strlen() and strev() using a C-style char array.

Find and replace a substring within a string.

### Program:

```
C++ Quest24.cpp > ...
1  #include <iostream>
2  #include <cctype>
3  using namespace std;
4
5  int main() {
6      char str[100];
7      int vowels = 0, consonants = 0;
8
9      cout << "Enter a string: ";
10     cin.getline(str, 100);
11
12     for (int i = 0; str[i] != '\0'; i++) {
13         char ch = tolower(str[i]);
14
15         if (ch >= 'a' && ch <= 'z') {
16             if (ch == 'a' || ch == 'e' || ch == 'i' ||
17                 ch == 'o' || ch == 'u') {
18                 vowels++;
19             } else {
20                 consonants++;
21             }
22         }
23     }
24
25     cout << "Vowels: " << vowels << endl;
26     cout << "Consonants: " << consonants << endl;
27
28     return 0;
29 }
30
```

### Output:

```
collect2.exe: error: ld returned 1 exit status
PS D:\Data(Structures)> cd "d:\Data(Structures)" ; if ($?) { g++ Quest24.cpp -o Quest24 } ; if ($?) { .\Quest24 }
Enter a string: hello
Vowels: 2
Consonants: 3
PS D:\Data(Structures)>
```

## Program:

```
... Quest20.cpp X Quest21.cpp X Quest22.cpp X Quest23.cpp X Quest24.cpp X Quest25.cpp X
C++ Quest25.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      char sentence[200];
6      int words = 0;
7      bool inWord = false;
8
9      cout << "Enter a sentence: ";
10     cin.getline(sentence, 200);
11
12     for (int i = 0; sentence[i] != '\0'; i++) {
13         if (sentence[i] != ' ' && sentence[i] != '\t') {
14             if (!inWord) {
15                 words++;
16                 inWord = true;
17             }
18         } else {
19             inWord = false;
20         }
21     }
22
23     cout << "Number of words: " << words << endl;
24
25     return 0;
26 }
27
```

## Output:

```
Problems Output Debug Console Terminal Ports
PS D:\Data(Structures)> cd "d:\Data(Structures)\\" ; if ($?) { g++ Quest25.cpp -o Quest25 } ; if ($?) { .\Quest25 }
Enter a sentence: hi my name is Anas khan
Number of words: 6
PS D:\Data(Structures)> |
```

## Program:

```
#include <iostream>
using namespace std;

// Our own strlen() function
int myStrlen(char str[]) {
    int length = 0;

    while (str[length] != '\0') {
        length++;
    }

    return length;
}

// Our own strrev() function
void myStrrev(char str[]) {
    int start = 0;
    int end = myStrlen(str) - 1;

    while (start < end) {
        char temp = str[start];
        str[start] = str[end];
        str[end] = temp;

        start++;
        end--;
    }
}
```

```
28 }
29
30 int main() {
31     char str[100];
32
33     cout << "Enter a string: ";
34     cin.getline(str, 100);
35
36     cout << "Length: " << myStrlen(str) << endl;
37
38     myStrrev(str);
39
40     cout << "Reversed string: " << str << endl;
41
42     return 0;
43 }
44
```

## Output:

```
Problems  Output  Debug Console  Terminal  Ports
PS D:\Data(Structures)> cd "d:\Data(Structures)\\" ; if ($?) { g++ Quest26.cpp -o Quest26 } ; if ($?) { .\Quest26 }
Enter a string: anasklhan
Length: 9
Reversed string: nahlskna
PS D:\Data(Structures)>
```

## Lab Exercises 1.9

### Task 1:

Write a function that uses a pointer parameter to double the value of a variable.

### Program:

```
Quest27.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  // Function receives the address of a variable
5  void doubleValue(int *ptr) {
6      *ptr = *ptr * 2; // Double the value stored at that address
7  }
8
9  int main() {
10     int number;
11
12     cout << "Enter a number: ";
13     cin >> number;
14
15     cout << "Before doubling: " << number << endl;
16
17     // Pass the address of number to the function
18     doubleValue(&number);
19
20     cout << "After doubling: " << number << endl;
21
22     return 0;
23 }
24
```

### Output:

```
Problems  Output  Debug Console  Terminal  Ports
PS D:\Data(Structures)> cd "d:\Data(Structures)\\" ; if ($?) { g++ Quest27.cpp -o Quest27 } ; if ($?) { .\Quest27 }
Enter a number: 2001
Before doubling: 2001
After doubling: 4002
PS D:\Data(Structures)>
```

### Task 2:

Demonstrate the difference between a pointer and a reference with code comments explaining each line.

## Program:

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      // ----- POINTER -----
7
8      int a = 10;          // Create an integer variable a
9      int *ptr = &a;       // Pointer stores the address of a
10
11      cout << "Value of a: " << a << endl;
12
13      cout << "Value using pointer: " << *ptr << endl;
14      // *ptr accesses the value stored at the address
15
16      *ptr = 20;           // Change a through the pointer
17
18      cout << "New value of a: " << a << endl;
19
20
21      // ----- REFERENCE -----
22
23      int b = 30;          // Create another integer variable b
24      int &ref = b;        // Reference becomes another name for b
25
26      cout << "Value of b: " << b << endl;
27
28      cout << "Value using reference: " << ref << endl;
29      // ref directly refers to b
30
31      ref = 40;            // Change b through the reference
32
33      cout << "New value of b: " << b << endl;
34
35      return 0;
36  }
```

## Output:

```
After building Quest28
PS D:\Data(Structures)> cd "d:\Data(Structures)\"; if ($?) { g++ Quest28.cpp -o Quest28 }; if ($?) { .\Quest28 }
Value of a: 10
Value using pointer: 10
New value of a: 20
Value of b: 30
Value using reference: 30
New value of b: 40
PS D:\Data(Structures)> 
```

## Task 3:

Create a dynamic 2D array using pointers, fill it, and free the memory correctly.

## Program:

```
C++ Quest29.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int rows, columns;
6
7      cout << "Enter number of rows: ";
8      cin >> rows;
9
10     cout << "Enter number of columns: ";
11     cin >> columns;
12
13     // Create an array of pointers
14     int **matrix = new int*[rows];
15
16     // Create an integer array for each row
17     for (int i = 0; i < rows; i++) {
18         matrix[i] = new int[columns];
19     }
20
21     // Fill the matrix
22     cout << "Enter elements of the matrix:" << endl;
23
24     for (int i = 0; i < rows; i++) {
25         for (int j = 0; j < columns; j++) {
26             cin >> matrix[i][j];
27         }
28     }
29
30     // Display the matrix
31     cout << "Matrix:" << endl;
```

```

for (int i = 0; i < rows; i++) {

    // Display the matrix
    cout << "Matrix:" << endl;

    for (int j = 0; j < columns; j++) {
        cout << matrix[i][j] << " ";
    }
    cout << endl;
}

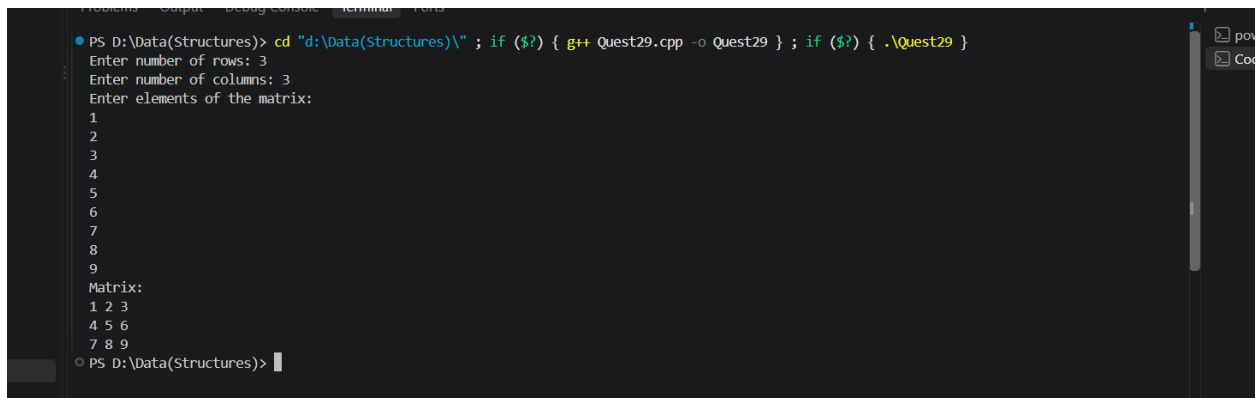
// Free each row
for (int i = 0; i < rows; i++) {
    delete[] matrix[i];
}

// Free the array of row pointers
delete[] matrix;

return 0;
}

```

## Output:



```

PS D:\Data(Structures)> cd "d:\Data(Structures)\\" ; if ($?) { g++ Quest29.cpp -o Quest29 } ; if ($?) { .\Quest29 }
Enter number of rows: 3
Enter number of columns: 3
Enter elements of the matrix:
1
2
3
4
5
6
7
8
9
Matrix:
1 2 3
4 5 6
7 8 9
PS D:\Data(Structures)>

```